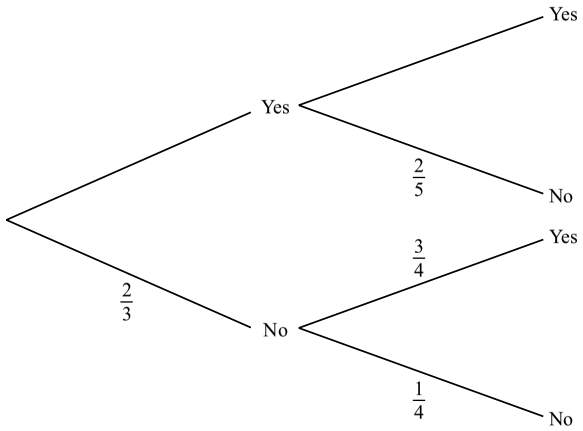
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | –2, –1, 0, 1 | **2** | **B1** if one error/omission/extra |
| 2 | Correct triangle with arcs | **2** | **B1** correct no arcs |
| 3 | $x^9y^4$ final answer | **2** | **B1** for $x^9y^k$ or $x^ky^4$ or correct answer seen then spoiled |
| 4 | –14, –5, 1 | **2** | **B1** for 2 correct |
| 5(a) | 32.7 to 33.3 | **1** | **FT** *their* printed diagram |
| 5(b) | 118 to 120 | **1** | **FT** *their* printed diagram |
| 5(c) | 298 to 300 | **2** | **B1** for 360 – (180 – *their* **(b)**)<br>**FT** *their* **(b)** + 180 |
| 6(a) | 5 | **2** | **M1** for 10 × 12 × 'h' = 600 |
| 6(b) | 172 or 172.4 to 172.5 | **2** | **M1** for $1120 \times \frac{4}{3} \times \pi \times 0.45^3$ |
| 7(a) | 20 | **1** | |
| 7(b) |  <br>$\frac{2}{3}, \frac{2}{5}, \frac{3}{4}, \frac{1}{4}$ oe | **2** | **M1** for 2 or 3 correct |
| 7(c) | $\frac{7}{10}$ oe | **3** | **M2** for $\frac{1}{3} \times \frac{3}{5}$ + *their* $\frac{2}{3}$ × *their* $\frac{3}{4}$<br>or **M1** for one product |

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| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(a) | $y = x$ ruled<br>$x + y = 7$ ruled | 3 | **B1** for $y = x$<br>**B2** for $x + y = 7$<br>or **B1** for ruled line with gradient –1 or ruled line through (0, 7) or (7, 0) but not $x = 7$ or $y = 7$. |
| 8(b) | Correct region below $y = x$, above $x + y = 7$ and above $x$-axis labelled R | 1 | |
| 9(a) | $\frac{8}{3}$ oe | 1 | |
| 9(b) | 92 | 2 | **M1** for correct calculation of an area |
| 10 | $(x + 3)(2 – 3y)$ | 2 | **M1** for $2(x + 3) – 3y(x + 3)$<br>or $x(2y – 3) + 3(2y – 3)$ |
| 11(a) | 6 | 2 | **M1** for $10 + 13 – (20 – 3)$ oe or correctly completed Venn diagram |
| 11(b) | 7 | 1 | **FT** *their* Venn diagram or 13 – *their* **(a)** |
| 12 | 160.375 | 4 | **M1** for mid-values for three of 125, 155, 162.5, 175<br>**M1** for $\Sigma fx$ where $x$ is in correct interval including boundary<br>**M1 dep** $\frac{\Sigma fx}{100}$ dep on second **M1** |
| 13(a) | 49.4 or 49.39 to 49.40 | 2 | **M1** for $\tan = \frac{14}{12}$ oe |
| 13(b) | 22.4 or 22.36 to 22.37 | 4 | **B1** for 62 + *their* **(a)**<br>**M2** for $\sqrt{12^2 + 15^2 - 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})}$<br>OR<br>**B1** for 62 + *their* **(a)**<br>**M1** for<br>$12^2 + 15^2 – 2 \times 12 \times 15 \cos(62 + \textit{their } \mathbf{(a)})$<br>**A1** for 500.3 to 500.4 |
| 13(c) | 13.2 or 13.24... | 3 | **M2** for $\sin 62 = \frac{\text{dist}}{15}$<br>or **M1** for recognition of shortest distance soi |

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